

Editor

Both reviewers agree the paper has potential, but the current version suffers from

- (i) conceptual overreach,
- (ii) methodological sprawl without sufficient justification/validation, and
- (iii) threats to inference from data representativeness, spatial dependence, and supply-side changes.

A major revision is warranted if the authors are willing to narrow the scope, tighten the modeling strategy, and add robustness/diagnostics that support their claims.

Reviewer 1:

1) Treating 2021 as a fully post-pandemic year may be problematic, given ongoing restrictions, behavioral adaptations, and uneven recovery across neighborhoods. The authors should justify this choice more carefully and/or conduct sensitivity analyses that either exclude 2021 or **treat it separately from a truly post-pandemic period**.

2) **A deeper discussion of potential sampling biases** in the SafeGraph data (e.g., overrepresentation of smartphone users with location tracking enabled) is needed. The authors should also include sensitivity analyses demonstrating that results are robust to concerns about representativeness across socioeconomic groups.

3) The estimated parameters are better interpreted as changes in revealed shopping patterns under changing constraints (including restrictions, closures, and mobility conditions), rather than as pure consumer preferences. The conceptual framing of the paper should be adjusted accordingly.

4) The use of a "top 5%" threshold for socioeconomic groups is somewhat arbitrary. The authors should consider more robust alternatives such as deciles, continuous regressions, or **a PCA-based clustering of CBGs** to capture socioeconomic heterogeneity more systematically.

5) Given that the paper presents spatial maps of CBG clusters, the authors should **test for spatial autocorrelation** (e.g., Moran's I and/or LISA) in residuals or in parameter changes. If spatial dependence is present, they should consider spatial models to ensure that neighborhood spillovers are not driving their results, or clearly acknowledge any remaining spatial dependence as a limitation of the study.

6) The authors should explicitly consider **how temporary or permanent closures of stores during COVID-19 -particularly smaller retailers-** may have shaped the observed mobility patterns. Some of the changes attributed to "consumer behavior" could in fact reflect changes in the availability of stores, rather than true preference shifts. The paper should discuss this more clearly and, where feasible, account for these supply-side changes in the analysis.

7) The visual presentation of Figures 2-7 should be improved. Several panels are difficult to read due to small fonts, overlapping labels, and inconsistent scales across subplots. The authors are encouraged to enlarge text, harmonize axis ranges where appropriate, and clarify legends to facilitate interpretation.

Reviewer 3

The paper aims to answer the question regarding whether COVID had any effects on people's shopping habits in NYC temporarily and whether their behaviour changed in a persistent manner. To answer that, the authors combine SafeGraph mobility data, POI/place data, and census block group demographics for New York City from 2018-2021. They estimate a modified gravity model of store choice and represent store choice using six factors: customer-store distance, store area, POI count near the store, POI diversity near the store, demographic similarity between the customer's CBG and the store's CBG, and chain loyalty. They normalize the attractiveness variables, calibrate the preference exponents with particle swarm optimization, and tune the normalization bounds with simulated annealing. After fitting yearly preference parameters, they compare each year 2018-2019, 2019-2020, and 2020-2021, and then validate the findings using store pair ground truth comparisons, K-means clustering, and Difference-in-Differences.

This leads them find that preferences in store selection did shift as seen in 2020, customers put more emphasis on larger stores, store locations and chain loyalty compared to the previous year with this behaviour persisting into 2021 instead of reverting back to the same patterns pre-COVID. They also show heterogeneity across socioeconomic groups and across K-means clusters built from combined demographic features. Overall, the paper is an interesting contribution and examines and shows evidence of persistent post-shock restructuring of urban retail choice.

1) The authors may find some interesting results if they choose to include an analysis of the stability under perturbation into the NYC retail setting. They could perturb the mobility matrix or fitted preference vectors and ask which retail choice clusters are stable, which retail "corridors" between neighborhoods and store districts are fragile, and which demographic communities exhibit structurally robust versus shock-sensitive destination preferences. This may strengthen the robustness since fitted correlations drop after 2019, from about 0.964 and 0.956 pre-pandemic to 0.742 and 0.544 in 2020 and 2021 so post-shock behavior may be noisier and harder to explain with one structure.

2) Another interesting point that the author may consider is the corridor resilience in the retail mobility system. Since the manuscript already models visits from residential CBGs to stores and calibrates these flows using observed visit distributions, it is possible to reinterpret the data as a network of shopping connections between origin neighborhoods and destination retail areas. This may allow the authors to move beyond changes in preference parameters alone and

examine which neighborhood-to-retail corridors remained robust after COVID-19 and which became fragile. This type of analysis can show that shopping preferences changed and that the spatial structure of retail access itself was reorganized, with some channels of mobility proving persistent and others vulnerable to disruption.

3) Some citations of interest which have aspects of these interesting additions that may be useful for the authors are given below.

Chang, S., Pierson, E., Koh, P.W. et al. Mobility network models of COVID-19 explain inequities and inform reopening. *Nature* 589, 82-87 (2021).

Mukhamedova, F., Tyukin, I. Machine learning analysis of the effects of COVID-19 on migration patterns. *Sci Rep* 14, 29815 (2024)

4) I think a minor addition would be helpful:

Since Section 5.3 already includes validation with ground truth, K-means heterogeneity analysis, and DID, a minor but very valuable addition would be a short stability check for the clustering. For example, rerun K-means under different seeds, or bootstrap the CBG sample and report whether the main cluster-level patterns remain. That would strengthen the robustness section.